

The nanoscale organization of the Nipah virus fusion protein informs new membrane fusion mechanisms

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Abstract

Paramyxovirus membrane fusion requires an attachment protein for receptor binding and a fusion protein for membrane fusion triggering. Nipah virus (NiV) attachment protein (G) binds to ephrinB2 or -B3 receptors, and fusion protein (F) mediates membrane fusion. NiV-F is a class I fusion protein and is activated by endosomal cleavage. The crystal structure of a soluble GCN4-decorated NiV-F shows a hexamer-of-trimer assembly. Here, we used single-molecule localization microscopy to quantify the NiV-F distribution and organization on cell and virus-like-particle membranes at a nanometer resolution. We found that NiV-F on biological membranes forms distinctive clusters that are independent of endosomal cleavage or expression levels. The sequestration of NiV-F into dense clusters favors membrane fusion triggering. The nano-distribution and organization of NiV-F are susceptible to mutations at the hexamer-of-trimer interface, and the putative oligomerization motif on the transmembrane domain. We also show that NiV-F nanoclusters are maintained by NiV-F-AP-2 interactions and the clathrin coat assembly. We propose that the organization of NiV-F into nanoclusters facilitates membrane fusion triggering by a mixed population of NiV-F molecules with varied degrees of cleavage and opportunities for interacting with the NiV-G/receptor complex. These observations provide insights into the in-situ organization and activation mechanisms of the NiV fusion machinery.

eLife assessment

This is a **valuable** paper that uses super-resolution microscopy to show the nanoclustering of the Nipah virus fusion protein on cell and viral membranes. Some of the conclusions regarding the clustering of viral fusion proteins is supported by **solid** biochemical and super-resolution imaging data while other conclusions such as significance for viral fusion mechanisms is not fully supported by the data provided.

Introductio

The family of *Paramyxoviridae* contains many clinically important viruses, such as human parainfluenza virus 1-3, measles virus, and Nipah virus (NiV). Paramyxoviruses code for two glycoproteins on the viral membranes for virus-cell membrane fusion. The receptor-binding proteins (RBP, HN/H/G) engage the host receptors and activate the refolding of the fusion protein (F) that merges the virus and cell membranes¹. Paramyxovirus-F is a class I fusion protein, along with human immunodeficiency virus (HIV) envelope, influenza virus hemagglutinin, and severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) spike¹. Paramyxovirus-F is a trimeric transmembrane protein with a “tree-like” structure at its prefusion conformation. Upon triggering by the RBP/receptor complex, paramyxovirus-F mediates fusion by inserting its N-terminal fusion peptides into the target membrane and undergoes a conformational change from pre-to post-fusion conformations, similarly to other class I fusion proteins. The refolding of the paramyxovirus-F overcomes the energy barrier for membrane fusion and leads to virus-cell and cell-cell membrane fusion.

NiV is a well-studied paramyxovirus due to its capability of human-human and human-animal transmissions, high mortality and morbidity rates, and the unavailability of vaccines or therapeutics for human use^{2,3}. NiV is closely related to Hendra virus, Cedar virus, and the newly identified Langya virus⁴. NiV-F is expressed on the cell surface as an fusion-inactive form, activated in the endosome by cathepsin B and L cleavage, and exocytosed to the cell surface as a fusion-active form^{5,6}. NiV-G binds to ephrinB2 and/or -B3 receptors in host cells and triggers the refolding of F that leads to virus-cell membrane fusion⁷. Three complementary models of the NiV fusion activation mechanism have been proposed in the past. *In the first model*, the NiV-G and F form a complex before receptor engagement. The F/G interaction maintains F at a prefusion conformation. The receptor binding to G induces conformational changes in G, leading to the release and refolding of F. This model is supported by co-immunoprecipitation and conformational antibody binding assays^{8,9}. *In the second model*, NiV-F exists as hexamer-of-trimers, and the fusion peptides are sequestered at the hexameric interface. The activation of one single trimer by the NiV-G/ephrinB2 complex can be transmitted to all six trimers via the interfaces of the hexamer-of-trimers, leading to efficient F-triggering and fusion pore formation. This model is supported by the crystal structure of a GCN4-decorated NiV-F ectodomain and mutagenesis analysis of key amino acid residues at the hexameric interfaces¹⁰. *In a third model*, NiV-F and G do not form a complex before receptor binding. The NiV-G/ephrinB2 engagement clusters ephrinB2 and triggers F. F triggering leads to reduced mobility of a portion of F molecules that potentially contributes to fusion pore formation. This model is supported by the nanoscale distribution of NiV-F and G on cell membranes resolved by super-resolution microscopy and a single-particle tracking assay of NiV fusion protein at the interface between supported lipid bilayer and live cell membranes^{11,12}. Although these models explain the NiV membrane fusion mechanisms from different perspectives, a link between the molecular structure and the fusion machinery on biological membranes remains unclear.

We attempted to address this link by analyzing the nano-organization of NiV-F at the prefusion state on biological membranes using single-molecule localization microscopy (SMLM) and mutagenesis analyses. We have shown that NiV-F forms clusters on biological membranes that are isolated from NiV-G¹¹. Here, we show that NiV-F forms regular-sized nanoclusters on cell membranes regardless of the expression level or endosomal cleavage. The estimated size of NiV-F clusters on cell membrane is similar to that of the hexamer-of-trimer assemblies formed by the GCN4-decorated soluble NiV-F. The NiV-F nano-organization is altered by mutations at the hexameric interface and the putative oligomerization motifs at the transmembrane domain. NiV-F molecules enriched in nanoclusters favor membrane fusion activation. The NiV-F nanoclusters are

stabilized by the interactions between NiV-F, the endocytosis adaptor complex AP-2, and the clathrin coat at the cell membranes. In summary, our study reveals the NiV-F nano-organization on the biological membranes and provides novel insights to the NiV fusion machinery *in situ*.

NiV-F forms regular-sized clusters that are not affected by the surface expression level

We recently published that NiV-F formed distinctive nanoclusters on the plasma membrane regardless of the presence of NiV-G or ephrinB2¹¹. To investigate the determinants of the NiV-F nano-organization, we used SMLM to probe the NiV-F at various conditions. SMLM generates super-resolution images with nanometer resolution by localizing blinks from individual fluorophores with a distance greater than the diffraction limit of light¹³. PK13 cells have a flat morphology that is suitable for SMLM imaging. PK13 cells express little endogenous ephrinB2 and -B3 receptors for NiV and thus can eliminate any potential receptor-dependent clustering. Cell surface NiV-F was detected using a FLAG tag in its extracellular domain¹¹. The FLAG tags were employed to detect NiV-F because of the availability of high affinity and specificity anti-FLAG antibodies for a high labeling density, which is vital for the reconstruction accuracy of SMLM images¹⁴. To rule out the effect of the FLAG tag on NiV-F clustering, we compared NiV-F clusters formed by NiV-F-FLAG to that of a NiV-F-HA construct. Both HA and FLAG tags were inserted after amino acid residue 104, right before the fusion peptide (Fig. S1A). Visual examination of the SMLM images reveals that NiV-F-FLAG and NiV-F-HA form similar clusters (Fig. S1B). The clustering tendency, indicated by Hopkins' index, is comparable between NiV-F-HA and NiV-F-FLAG (Fig. S1C), suggesting that NiV-F clustering is not affected by specific epitope tags.

Next, we investigated whether the nano-organization of NiV-F depends on the cell surface expression levels. The total cell surface expression level (CSE) of NiV-F in a selected cell was measured before the cell was subjected to SMLM imaging. The SMLM images show that NiV-F forms similar nanoclusters in both high-expression and low-expression cells (Fig. 1A and B).

Our data show that the clustering tendency of NiV-F is independent of its CSE in individual cells (Fig. 1C). To gain a quantitative understanding of NiV-F nanoscale organization, we used a cluster identification method DBSCAN (Density-Based Spatial Clustering of Applications with Noise) that links the closely situated localizations in a propagative manner. We created cluster and density maps using Clus-DoC for representative high-expression and low-expression cells (Fig. 1D and E)¹⁵. Our data show that the estimated diameter of the NiV-F clusters ranges from 20-34 nm, with a peak at 24 nm (Fig. 1F). Nonetheless, the precise values of the cluster diameter are indicative rather than definite in SMLM as it could be affected by the labeling complex and the blinking property of the fluorophore. The distribution of the cluster diameter shows that ~60% of NiV-F clusters have similar sizes, indicating that NiV-F clusters maintain a quite uniform appearance. Interestingly, we noticed that the size of the clusters (Fig. 1G) and localization density (Fig. 1H) in clusters were similar between high- and low-expression cells, suggesting that the NiV-F organization in nanoclusters is not affected by the expression levels. Moreover, our data show that the high-expression cells have more clusters than the low-expression cells (Fig. 1I). These results suggest that changes in CSE may affect the number of NiV-F clusters on the plasma membrane (Fig. 1I), but not the clustering tendency of NiV-F (Fig. 1C), the size of the clusters (Fig. 1G), or the localization density within clusters (Fig. 1H). These data imply that the distribution and nano-organization of NiV-F are tightly regulated and may be important for membrane fusion activation. We also analyzed NiV-F distribution in different cell lines. HeLa cells express ephrinB2 and -B3, and thus are NiV permissive cells. The overall NiV-F nano-organization seems quite similar on PK13 and HeLa cells, agreeing with our previous results (Fig. S2A). The Hopkins' indices are also similar for NiV-F on PK13 and HeLa cells (Fig. S2B). The DBSCAN analysis shows that the percentage localizations in clusters (Fig. S2C), the relative density defined as the ratio of the localization density in clusters to the total localization density in the selected region

(Fig. S2D), and cluster diameters (Fig. S2E) are comparable for NiV-F on PK13 and HeLa cells. These results suggest that NiV-F clusters are minimally affected by the cell lines of expression or the presence of the ephrinB2 and/or -B3.

The NiV-F nanoclusters are not affected by NiV-F cleavage

NiV-F is synthesized in host cells as inactive precursor F_0 and cleaved by cellular proteases cathepsin L and cathepsin B in the endosomes at an acidic pH to generate the fusion-active, disulfide-linked F_1 - F_2 construct. The F_1 - F_2 is subsequently recycled to the cell surface to induce cell-cell fusion and incorporated into virus particles. We hypothesized that the F_0 precursor and the F_1 - F_2 active forms co-existed in the same clusters on the cell surface. To test this hypothesis, we used a pan-cysteine cathepsin inhibitor E64d to inhibit the NiV-F cleavage. E64d inhibited the cell-cell fusion induced by NiV-F and NiV-G in HeLa cells (Fig. S3), agreeing with the previous fusion inhibitory effects in Vero and MDCK cells^{5,6}. Our SMLM images show that E64d treatment does not result in any significant change in the clustering of NiV-F in HeLa cells (Fig. 2A). The clustering extent of NiV-F was not altered by the E64d treatment, as shown by comparable Hopkin's indices (Fig. 2B). Additionally, quantitative parameters of the nano-organization such as percentage of localizations in clusters (Fig. 2C), relative density (Fig. 2D), cluster size (Fig. 2E), and the total density of the regions (Fig. 2F). Our data show that clusters formed by non-cleaved NiV-F in cells treated by E64d are similar to those formed by a mixture of cleaved and non-cleaved NiV-F in the control cells, suggesting that F cleavage does not alter F clustering on the cell membrane. These data support our hypothesis that the non-cleaved precursor F_0 and cleaved F_1 - F_2 coexist in the same cluster on the cell membrane.

Mutations destabilizing the NiV-F hexamer-of-trimer assembly alter its nano-organization on plasma membrane

We noticed that the uniformity in the NiV-F cluster morphology resembled the hexameric assembly of soluble NiV-F decorated by GCN4¹⁰. Next, we investigated whether NiV-F clusters were stabilized by the hexameric interface. As reported, L53D and V108D mutants destabilize the hexameric interface and demonstrate decreased membrane fusion ability; Q393L stabilizes the hexameric interface and demonstrates increased fusion ability. We inserted the FLAG tag into the ectodomains of these mutants at the same position as that of the NiV-F. We verified that the relative fusion ability to their F-WT counterpart was comparable between the FLAG-tagged and untagged mutants (Fig. S4A and S4B)¹⁰. The FLAG-tagged mutants showed similar CSE levels to NiV-F (Fig. S4C). All mutants showed some levels of processing, although the cleavage of L53D was the least efficient comparing to F-WT and other mutants (Fig. S4D).

NiV-F mutants L53D, V108D, and Q393L were expressed on PK13 cells and subjected to SMLM imaging. The images show that clusters formed by L53D and V108D are smaller and more dispersed than those of the F-WT and Q393L (Fig. 3A). The Hopkin's index for Q393L is significantly higher than that of F-WT, while the Hopkin's indices for L53D and V108D are slightly lower than that of F-WT (Fig. 3B). Consistently, a lower percentage of L53D and V108D localizations are segregated into clusters than that of F-WT, while a similar percentage of Q393L and F-WT localizations form clusters (Fig. 3C). Our analysis also shows that L53D and V108D form smaller clusters compared to the F-WT. The Q393L clusters are of a similar size to that of F-WT (Fig. 3D). In addition, clusters of L53D and V108D are less packed than that of F-WT, with 48-, 49-, and 55-fold denser than the total localization density in the region, respectively (Fig. 3E). The clusters of Q393L are not significantly denser than that of F-WT (Fig. 3E). The total density of localizations and the total CSE levels for all constructs are comparable (Fig. 3F and Fig. S4C), suggesting that differences in clustering are not due to variable numbers of localizations or levels of expression. Collectively, these data show that the mutations destabilizing the hexameric interface (L53D and V108D) make NiV-F localizations more dispersed and form smaller and less packed clusters. Q393L that stabilizes the hexameric interface promotes the clustering of

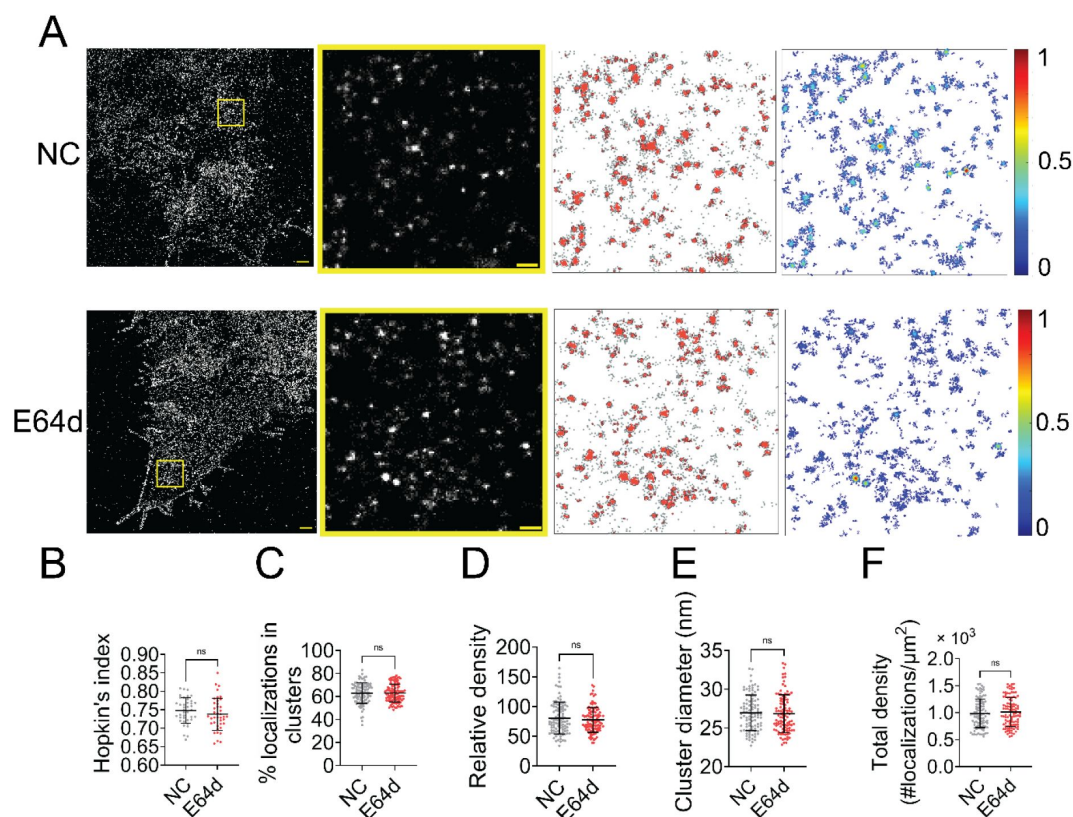


Fig. 2.

Endosomal cleavage does not affect the nanoscale distribution of NiV-F.

(A) First column: Cross-section ($\Delta z = 600$ nm) of SMLM images of NiV-F in HeLa cells untreated (NC) and treated with 20 μ M E64d (E64d). HeLa cells were co-transfected by expression plasmids coding for NiV-G and NiV-F. Twenty μ M E64d or the same volume of solvent methanol was added to cells at 2 hrs post-transfection. Scale bar: 1 μ m. Second column: The yellow boxed region in the first column is enlarged to show individual clusters. Scale bar: 0.2 μ m. Third column: Cluster maps from the enlarged regions. Fourth column: Localization density maps show the normalized relative density of the enlarged regions. (B-F) Quantitative analyses of NiV-F clusters formed in HeLa cells without (NC) and with (E64d) the E64d treatment: (B) Hopkin's index, $n = 40$ and 40 ; (C) percentage of localizations in clusters, $n = 101$ and 101 ; (D) relative density, $n = 100$ and 103 ; (E) average cluster diameters, $n = 100$ and 101 ; (F) total density of the region, $n = 101$ and 102 . Bars represent mean $\pm$ SD. p value was obtained using Mann-Whitney test. ns: $p > 0.05$; * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$. Sample size n is the number of total regions from 4-10 cells.

NiV-F localizations (**Fig. 3B**), but does not significantly alter the organization and morphology of individual clusters (**Fig. 3C-F**). These data suggest that the NiV-F clusters are susceptible to mutations at the hexameric interface. Considering that the uniformed cluster size is similar to the assemblies formed by the soluble GCN4-NiV-F, we propose that NiV-F may organize into hexamer-of-trimer on the plasma membrane.

Mutations at the NiV-F hexameric interface affect its distribution on the VLP membrane

To further investigate the organization of F mutants in viral membranes, we imaged the FLAG-tagged NiV-F constructs on VLPs produced by 293T cells expressing NiV-M-GFP, NiV-G, and NiV-F-WT, L53D, V108D, or Q393L mutants. The western blot analysis of NiV-F constructs on the VLPs (**Fig. 4A**). The L53D and V108D on VLPs are less cleaved compared to that of the F-WT and Q393L, as suggested by the weaker NiV-F₂ bands (**Fig. 4A**). The incorporation of NiV-F is comparable among the NiV-F constructs (**Fig. 4A**). Notably, the incorporation of NiV-G is largely abrogated in the VLPs that expressing L53D (**Fig. 4A**).

To probe the organization of NiV-F constructs on VLPs, the F-constructs on VLP membranes were stained using Alexa Fluor 647 and subjected to SMLM imaging. The GFP fluorescence on NiV-M was used to locate the VLPs. **Fig. 4B** shows that F-WT and F-Q393L form distinctive clusters, while localizations of L53D and V108D are more dispersed on a $z = 100$ nm projection. This agrees with their distributions on the plasma membrane (**Fig. 3A**). To gain a quantitative insight into the distribution of the F constructs on the three-dimensional (3D) VLPs, we used an algorithm named OPTICS (Ordering Points To Identify Clustering Structure)¹⁶. To identify clusters, the OPTICS algorithm sorts out the localizations by calculating the reachability distance between two adjacent localizations in a propagative manner and generates a sequence of the ordered localizations¹⁶. In the plot of reachability distance vs. the sequence of ordered localizations, a sudden increase in the reachability distance marks the cutoff of a cluster (**Fig. S5A**). The shade between neighboring peaks represents localizations that are classified in one cluster (**Fig. S5A**). OPTICS can identify sub-clusters on a 3D VLP that may not be recognized by DBSCAN that uses one fixed parameter for the entire dataset. **Fig. S5B** shows the OPTICS plots of the localizations of F-WT and mutants on representative VLPs (**Fig. 4B**).

The noticeable kinks suggest that F-WT and Q393L form distinctive clusters on VLPs, while the irregular, small kinks indicate that L53D and V108D are more dispersed and form smaller clusters on VLPs (**Fig. 4B** and **Fig. S5B**). To gain a population insight on the nano-organization of the F constructs on VLPs, we developed a one-dimensional convolutional neural network (1D CNN) to classify the ordered sequence of reachability distances of the NiV-F localizations obtained by the OPTICS algorithm¹⁷. The distribution patterns of the localizations of the F-WT and Q393L partition into one category, and L53D and V108D in another (**Fig. 4C**). These results indicate that L53D and V108D form smaller and more dispersed clusters than F-WT and Q393L on the VLP membranes, agreeing with that of the plasma membrane (**Fig. 3**). Our results indicate that the trimer-trimer interaction at the hexameric interface is key in stabilizing the nano-organization of NiV-F on both cell and viral membranes, and NiV-F do not seem to rearrange during the incorporation into VLPs. Nonetheless, our data do not rule out the possibility that the association of nucleocapsids with the plasma membrane during assembly may reorganize NiV-F on the authentic virus membranes.

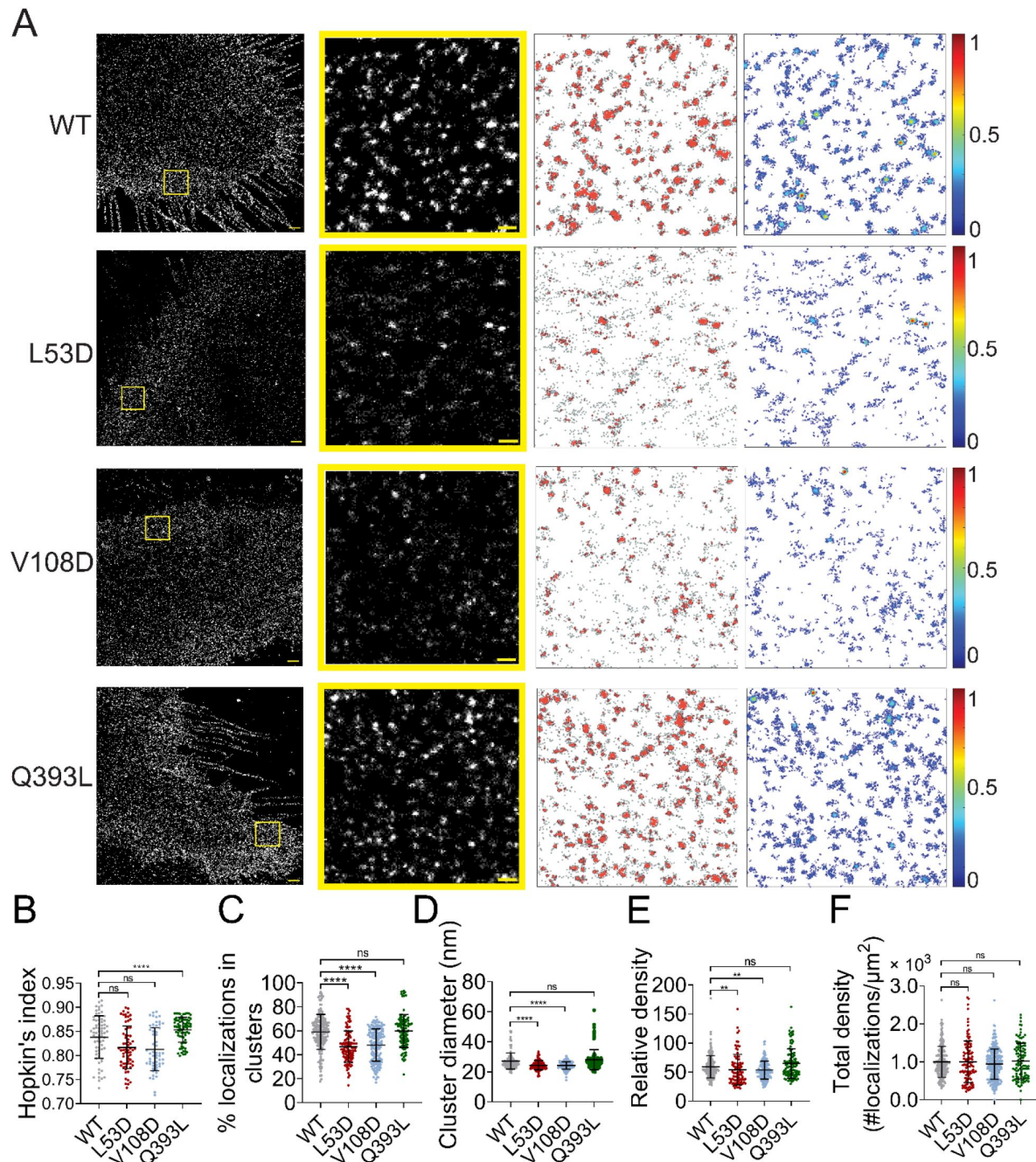


Fig. 3.

Mutations at the NiV-F hexameric interface affect its nano-organization.

(A) First column: Cross-section ($\Delta z = 600$ nm) of SMLM images of the FLAG-tagged NiV-F-WT (WT), L53D, V108D, and Q393L on PK13 cell membrane. Scale bar: 1 μm . Second column: The yellow boxed regions in the first column are enlarged to show individual clusters. Scale bar: 0.2 μm . Third row: Cluster maps from enlarged regions. Fourth column: Localization density maps show normalized relative density of the enlarged regions. (B-F) Quantitative analyses of the distribution of the FLAG-tagged NiV-F constructs: (B) Hopkin's index, $n = 57-70$; (C) Percentage of localizations in clusters, $n = 106-198$; (D) Average cluster diameters, $n = 106-196$; (E) Relative density, $n = 90-187$; (F) total density of the region (a ratio of total localizations in a region to the size of the region), $n = 105-242$. Bars represent mean $\pm$ SD. Sample size n is the number of total regions from 11-20 cells. p value was obtained using Mann-Whitney test. ns: $p > 0.05$; * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$.

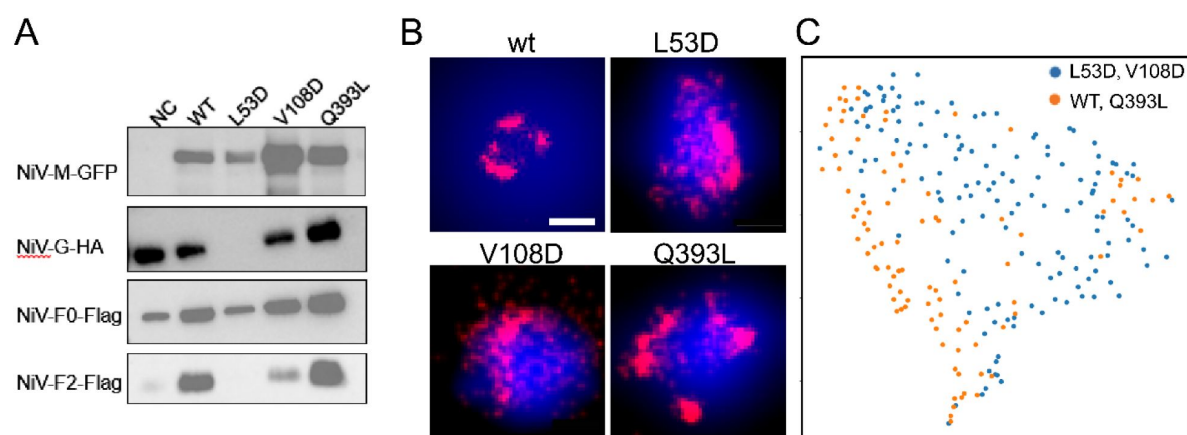


Fig. 4.

The distribution and organization of NiV-F constructs in VLPs.

(A). The incorporation of F-WT and mutants in VLPs. NiV-M-GFP, G-HA, and FLAG-tagged F-WT or mutants were transfected to 293T cells. The supernatants were collected at 48 hrs post-transfection and analyzed on SDS-PAGE followed by western blotting. NiV-M was probed by polyclonal goat anti-GFP, NiV-G polyclonal rabbit anti-HA, F₀ and F₂ M2 monoclonal mouse anti-FLAG antibody. (B) Cross-section ($\Delta z = 100$ nm) of SMLM images of the FLAG-tagged NiV-F-WT (WT), L53D, V108D, and Q393L on individual VLPs. Scale bar: 0.2 μ m. (C) The classification of the ordered sequence of reachability distances of the NiV-F constructs localizations. Orange: F-WT and Q393L; Blue: L53D and V108D.

The NiV-F clusters are dispersed upon the disruption of the interactions between transmembrane domains (TMD)

Evidence shows that TMD of class I fusion protein can self-associate in the absence of the rest of the protein and is important for membrane fusion¹⁸. Previous studies show that mutations in the Leucine-Isoleucine Zipper (LI zipper) motif lead to the dissociation of an HeV-F TMD-derived peptide, decreased stability, and the fusion ability of the whole HeV-F protein¹⁹. As the TMD domains of the NiV- and HeV-F demonstrate a 94% similarity, we mutated leucine (488 and 509) and isoleucine (495 and 502) residues of the LI zipper in NiV-F TMD to alanine (LI4A) (Fig. S6A). Similar to their counterparts of HeV-F, the NiV-F-LI4A showed a decreased F processing (Fig. S6B), cell-cell fusion activity (Fig. S6C and S6D), and CSE levels (Fig. S6E)¹⁹. Visual inspection of the images suggests that LI4A forms bigger clusters than F-WT (Fig. 5A). However, the LI4A seems less likely to cluster than F-WT, indicated by a lower Hopkin's index (Fig. 5B). Similarly, the portion of LI4A localizations partitioned into clusters is slightly lower than that of WT (Fig. 5C). Interestingly, we noticed that clusters formed by LI4A are significantly bigger (Fig. 5D) than that of F-WT and with a similar localization density within the clusters (Fig. 5E). The data suggest that interactions between TMDs of NiV-F monomers play a role in stabilizing its nano-organization. Dissociation of TMDs by mutations may result in increased space between F monomers and thus leads to bigger clusters on the cell membrane.

The NiV-F nanoclusters are stabilized by endocytosis components

NiV-F can be endocytosed for endosomal cleavage. Next, we investigated whether NiV-F clusters are stabilized during endocytosis. NiV-F contains an endocytosis sorting signal YSRL and an additional YY motif at its cytoplasmic tail (Fig. S7A). Mutation of these tyrosine residues to alanine almost diminished NiV-F cleavage (Diederich et al., 2005). A FLAG-tagged NiV-F-YA construct containing the aforementioned mutations resulted in significantly less F₂ band than F-WT (Fig. S7B) and completely abrogated cell-cell fusion in 293T cells (Fig. S7C and S7D), although the CSE levels were comparable to that of F-WT (Fig. S7E). By visually inspecting the SMLM images, we noticed that F-YA was less clustered than that of F-WT (Fig. 6A). Indeed, the Hopkin's analysis suggests that F-YA is less clustered than F-WT (Fig. 6B). The DBSCAN analysis shows that a lower percentage of F-YA localizations segregate into clusters (Fig. 6C) than that of the F-WT, and clusters formed by F-YA are smaller (Fig. 6D) and less dense (Fig. 6E) than that of the F-WT. Notably, there is no significant difference in the total density for F-YA and F-WT SMLM images (Fig. 6F), indicating that the differences in cluster organizations do not result from overall protein expression or the stochastic blinking properties of the fluorophore.

These results suggest that the endocytosis sorting signal of NiV-F may facilitate the enrichment of NiV-F into clusters on the cell surface, preparing them for endosomal cleavage and/or incorporation into viruses.

It is established that the YxxΦ sorting signal on the endocytosis cargo is bound and enriched by the heterotetrameric adaptor protein (AP) complexes, which further recruit the assembly of the clathrin coat^{20,21}. We hypothesize that the clustering of NiV-F is promoted by the assembly of the clathrin coat via the NiV-F-AP-2 interactions. As suggested by a yeast two-hybrid analysis, NiV-F interacts with μ1, μ2, μ3 and μ4 subunits of the AP complexes AP-1, AP-2, AP-3, and AP-4, and mutations of Y525, Y542, and Y543 almost completely abolished the interaction between AP-2 and NiV-F²². Indeed, the co-immunoprecipitation assay shows that less AP2μ2 is pulled down by F-YA than that of F-WT, suggesting that F-YA has a reduced interaction with AP-2 (Fig. S7F). Next, we examined the NiV-F clusters in HeLa cells treated by the endocytosis inhibitor pitstop2. Pitstop2 blocks clathrin-mediated endocytosis by obstructing the binding of the accessory protein (e.g. AP-2) and the clathrin terminal domain, and potentially preventing the assembly of the clathrin coat²³. The SMLM images (Fig. 6G) show that the NiV-F localizations are more dispersed upon

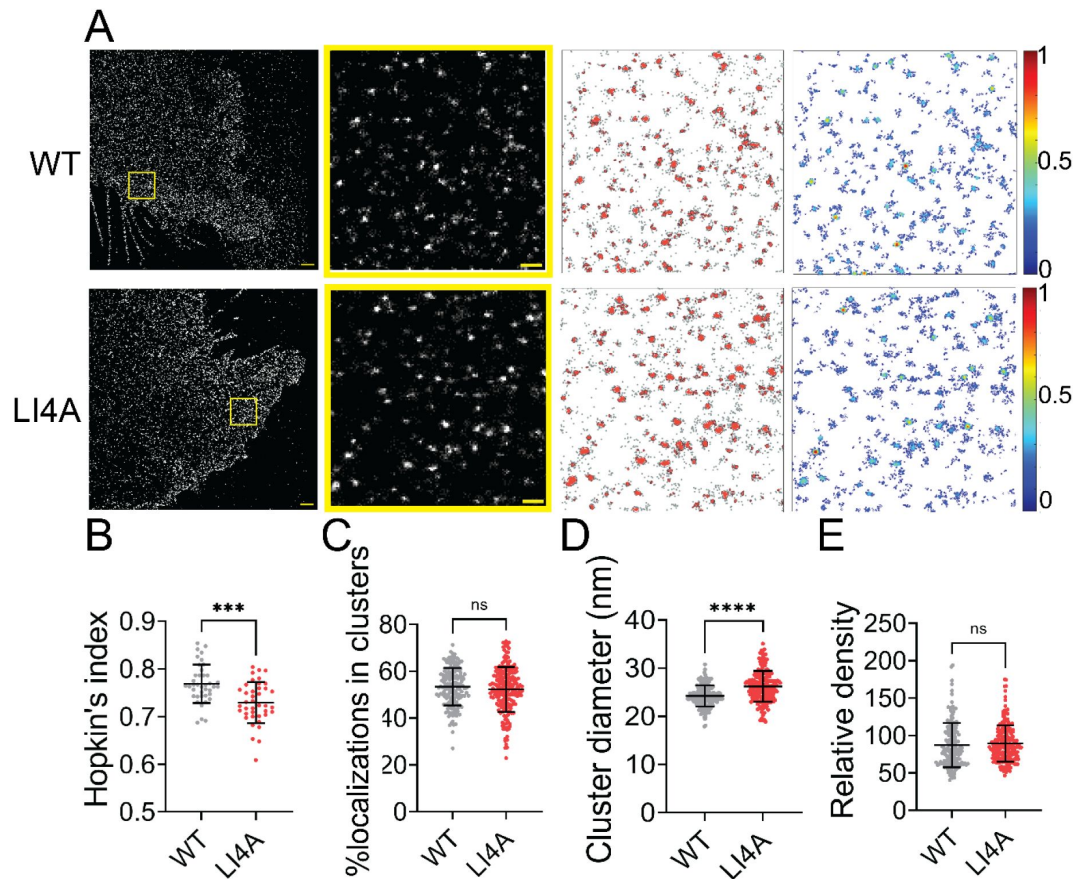


Fig. 5.

Mutations in the LI zipper of the NiV-F transmembrane domain disturbs the NiV-F distribution.

A) First column: Cross-section ($\Delta z = 600$ nm) of SMLM images of the FLAG-tagged NiV-F-WT (WT) and NiV-F-LI4A (LI4A) mutant on PK13 cell membrane. Scale bar: 1 μ m. Second column: The yellow boxed region in the first column is enlarged to show individual clusters. Scale bar: 0.2 μ m. Third column: Cluster maps from enlarged regions. Fourth column: Localization density maps show normalized relative density of the enlarged regions in the second row. (B-F) Quantitative analyses of the WT and LI4A clusters: (B) Hopkin's index, $n=40$ and 40 (C) Percentage of localizations in clusters, $n = 171$ and 211; (D) Average cluster diameters, $n = 171$ and 211; (E) Relative density, $n = 166$ and 211. Bars represent mean $\pm$ SD. p value was obtained using Mann-Whitney test. ns: $p > 0.05$; * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$. Sample size n is the number of total regions from 13-16 cells.

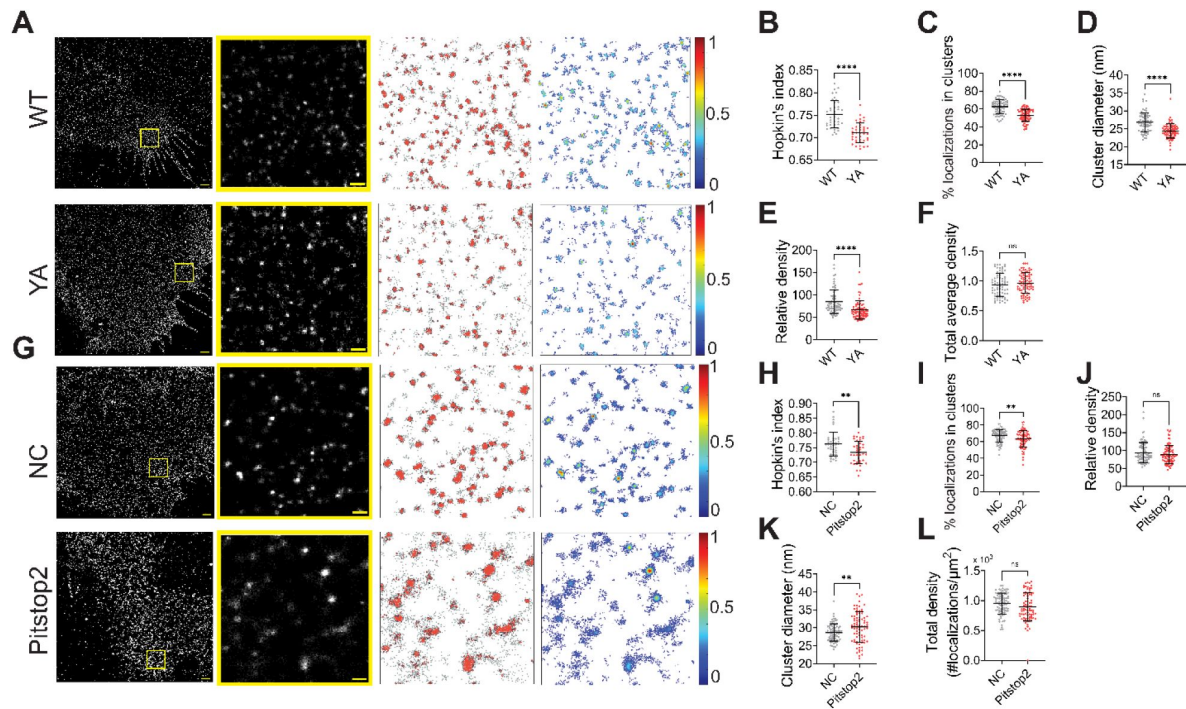


Fig. 6.

The NiV-F nanoclusters are stabilized by endocytosis components.

(A) First column: Cross-section ($\Delta z = 600$ nm) of SMLM images of the FLAG-tagged NiV-F-WT (WT) and NiV-F-YA (YA) mutant on PK13 cell membrane. Scale bar: 1 μm . Second column: The yellow boxed region in the first column is enlarged to show individual clusters. Scale bar: 0.2 μm . Third column: Cluster maps from enlarged regions. Fourth column: Localization density maps show normalized relative density of the enlarged regions in the second row. (B-F) Quantitative analyses of the WT and YA clusters: (B) Hopkins index of WT and YA, $n = 40$ and $n = 40$; (C) Percentage of localizations in clusters, $n = 72$ and 77 ; (D) Average cluster diameters, $n = 72$ and 77 ; (E) Relative density, $n = 71$ and 76 ; (F) total density of the region, $n = 72$ and 77 . (G) First column: Cross-section ($\Delta z = 600$ nm) of SMLM images of the FLAG-tagged NiV-F-WT treated without (NC) and with pitstop2 (pitstop2) on HeLa cell membrane. Scale bar: 1 μm . Second column: The yellow boxed region in the first column is enlarged to show individual clusters. Scale bar: 0.2 μm . Third column: Cluster maps from enlarged regions. Fourth column: Localization density maps show normalized relative density of the enlarged regions in the second row. (H-L) Quantitative analyses of NiV-F without (NC) and with pitstop2 (pitstop2): (H) Hopkins index, $n = 40$ and $n = 40$; (I) Percentage of localizations in clusters, $n = 82$ and 85 ; (J) Relative density, $n = 82$ and 81 ; (K) Average cluster diameters, $n = 82$ and 81 ; (L) total density of the region, $n = 82$ and 81 . Bars represent mean $\pm$ SD. p value was obtained using Mann-Whitney test. ns: $p > 0.05$; * $p < 0.01$; ** $p < 0.001$; *** $p < 0.0001$. Sample size n is the number of total regions from 4-9 cells.

pitstop2 treatment, agreeing with the lower Hopkin's index (**Fig. 6H**) and a smaller percentage of localizations in clusters (**Fig. 6I**) in pitstop2 treated group compared to that of the control group. Interestingly, we noticed that the NiV-F clusters in pitstop2 treated cells were larger than the control cells (**Fig. 6K**), although with similar densities (**Fig. 6J**). The total density is consistent between the pitstop2 treated and control groups (**Fig. 6L**). The overall dispersed NiV-F distribution upon pitstop2 treatment may potentially be caused by the disrupted assembly of clathrin coat. In combination, these results suggest that the enrichment of NiV-F by AP-2 and the subsequent assembly of the clathrin-coated endosomes are important in stabilizing the F clusters on cell membranes. We envision that the endocytosis components may also facilitate the NiV-F clustering in virus particles as previous proteomic studies show the presence of endocytosis related proteins, such as clathrin, AP-2, and dynamin in NiV VLPs^{24–26}.

Discussion

Here we used SMLM to resolve the arrangement and organization of NiV-F on the biological membranes at a resolution of 10 nm. Our data supports the following conclusions: 1) NiV-F is organized into nanoclusters on the biological membranes and this organization is independent of the protein expression level or endosomal cleavage; 2) the NiV-F nano-organization is susceptible to mutations at the trimer interface on the NiV-F ectodomain and the putative oligomerization motif on the transmembrane domain; 3) NiV-F sequestered in nanoclusters favors membrane fusion activation; 4) the interactions among NiV-F, the AP-2 complex, and the clathrin coat assembly stabilize the NiV-F clusters. We propose that the NiV-F nanoclusters are the fundamental unit of the NiV fusion machinery, and this organization facilitates membrane fusion triggering by a mixed population of NiV-F molecules with varied degrees of cleavage and opportunities of interacting with NiV-G/receptor complex.

The nano-organization of NiV-F informs the coordinated membrane fusion triggering by a mixed population of NiV-F. This notion is supported by the following observations from this and previous studies: 1) NiV-F molecules form distinctive, regular-sized nanoclusters on the cell and virus membranes (**Fig. 1A-E**); 2) the cleaved, active F co-exist with the noncleaved, inactive F in a nanocluster, as suggested by that the NiV-F nanoclusters are resistant to the cathepsin inhibitor, E64d (**Fig. 2**); 3) the NiV-F and G are segregated into different clusters¹¹; and 4) the full-length NiV-F and -G proteins do not show stable interactions either before or after ephrinB2 activation¹². These observations indicate that the NiV-F triggering may be a result of transient interactions between the F molecules at the edge of the clusters with the adjacent ephrinB2/G complex. Moreover, we observed that the NiV-F sequestering into clusters is favorable for membrane fusion activation (**Fig. 3** and **5**), highlighting the importance of the spatial arrangement of NiV-F in its fusion activity. Therefore, it is likely that the triggered, fusion-active NiV-F at the cluster peripheral can facilitate other F molecules in the same cluster to fulfill the conformational changes, and thus maximize the energy to merge the opposing membranes.

The nano-organization of NiV-F reveals the fundamental unit of NiV fusion machinery. Previous study revealed a hexamer-of-trimer assembly of soluble, GCN4-decorated, prefusion NiV-F. Using SMLM, we estimated that the NiV-F clusters had a diameter between 24–26 nm (**Fig. 1F**). This estimation agrees with the size of the hexameric soluble NiV-F-GCN4 assemblies¹⁰.

Notably, the density of clusters (**Fig. 1I**), but not the size and localization density of individual clusters (**Fig. 1G** and **H**), is affected by the surface expression levels of NiV-F, implying that the NiV-F clusters are highly regulated and a fundamental functional unit. Nonetheless, a collaborative effort of multiple copies of viral envelope proteins is favorable for overcoming the energy barrier required for membrane fusion. Nanoclusters formed by viral glycoproteins have been observed for HIV-1^{27,28}, Herpes virus²⁹, and Influenza virus³⁰. The viral restriction factor, serine incorporator protein 5 (SERINC5), inhibits HIV-1 fusion by disrupting the HIV env

nanoclusters, highlighting the importance of the nanoclusters in the function of the viral glycoproteins³¹. More interestingly, the NiV-F distribution becomes more dispersed, characterized by smaller and less dense clusters, upon the mutation of two key residues, L53 and V108, at the hexameric interface, suggesting that the NiV-F clusters are likely to be the hexamer-of-trimer assemblies on the cell and viral membranes (**Fig. 3** and **4**).

We also identified that the endosomal components as key host factors in maintaining the NiV-F nano-organization on the cell membrane. Our data suggest that both the disruption of the F/AP-2 interaction and the inhibition of clathrin assembly by pitstop2 result in dispersed NiV-F distribution. AP-2 is the key adaptor of clathrin-mediated endocytosis. It binds cargo and PtdIns(4,5)P₂-containing membrane via multiple interface, and undergoes conformational changes to append to the clathrin lattice^{23,32}. Our data indicate that AP-2-NiV-F interaction can enrich and strengthen the F nanoclusters (**Fig. 6A-F** and Fig. S7F), and potentially facilitate the uptake of NiV-F by endocytosis. It is reported that the clustering of CD44, modulated by N-glycosylations, facilitates the uptake of CD44 by endocytosis. Furthermore, the assembly of the clathrin coat at the plasma membrane may also strengthen the F clusters, because pitstop2 that inhibits the clathrin-coat assembly by blocking the clathrin terminal domain²³ prevents the cluster formation of NiV-F. It is plausible that both clathrin and AP-2 facilitate NiV-F clustering on cell and virus membranes because of the similar organization of F on virus and cell membranes and the presence of clathrin and AP-2 in VLPs^{24,26}.

In conclusion, our observations provide direct evidence on the nano-organization and distribution of NiV-F on the cell and viral membranes and shed lights to the fusion activation mechanisms. It would be interesting to analyze the reorganization of the NiV-F clusters upon triggering by the NiV-G/receptor complex on live cell membranes by combining single-molecule imaging and the supported lipid bilayer technologies¹². Due to technical challenges, we cannot obtain an exact number of NiV-F molecules in NiV-F clusters. It would be interesting to pinpoint the numbers of NiV-F required for membrane fusion activation on cell and virus surface, and this could potentially facilitate the vaccine design. Additionally, elucidating host factors that maintain NiV-F clusters and the interacting motif in NiV-F may represent a new therapeutic strategy for NiV infections.

Materials and Methods

DNA plasmid	NiV-M-GFP	Liu et al. 2018, Wang et al. 2010 11,33		
DNA plasmid	AP2μ2-mCherry	Christien Merrifield	Addgene 27672	
Chemical compounds, drug	E64d	Sigma-Aldrich	E8640-250UG	20uM
Chemical compounds, drug	Pitstop2	Sigma-Aldrich	SML1169-5MG	30uM
Antibody	Anti-FLAG mouse monoclonal	Sigma-Aldrich	F1804	IF, SMLM: 1:100, Flow: 1:200;
Antibody	Anti-HA rabbit polyclonal antibody	Biolegend	902301	WB: 1:500-1:5,000 IF: 1:900 WB: 1:2,000
Antibody	Anti-GFP goat polyclonal antibody	Abcam	Ab5450	WB: 1:1000
Antibody	Anti-mCherry rabbit polyclonal antibody	Abcam	Ab167453	WB: 1:2000
Antibody	Anti-mouse donkey polyclonal antibody, Alexa Fluor® 647 conjugated	Invitrogen	A31571	IF and SMLM: 1:400
Antibody	Anti-rabbit donkey polyclonal antibody, Alexa Fluor® 488 conjugated	Invitrogen	A21206	IF: 1:400
Antibody	Anti-mouse donkey polyclonal antibody, Alexa Fluor® 488 conjugated	Invitrogen	A21202	Flow: 1:400
Antibody	Anti-goat donkey polyclonal antibody, HRP conjugated	Jackson Immunoresearch	705-035-147	WB: 1:5,000
Antibody	Anti-mouse goat polyclonal antibody, HRP conjugated	Biorad	1705047	WB: 1:5,000

Antibody	Anti-rabbit goat polyclonal antibody, HRP conjugated	Biorad	1706515	WB: 1:5,000
Commercial assays and kits	µMACS anti-DYKDDDDK starting kit	Miltenyi Biotec	130-101-636	
software	SMLM image reconstruction	Liu et al. 2018 ¹¹		MATLAB Codes are available upon request. Codes were modified to process the data generated in the custom-built SMLM microscope. C++ Codes are available upon request. PYTHON Codes are available upon request.
software	ClusDoc	Pageon et al. 2015 ¹⁵		
software	OPTICS ¹⁶	This paper		
software	One-dimensional convolutional neural network (1D CNN)	This paper		
software	Prism	Dotmatics		
software	Graphpad			
software	Flowjo	BD		

Cell lines and plasmids

PK13, HeLa, and HEK293T cells were cultured at 37°C and 5% CO₂ in DMEM (Sigma-Aldrich, D6429) complemented with 10% fetal bovine serum (Invitrogen, 12483-020). Cells were passaged using phosphate-buffered saline (PBS, Invitrogen, 10010-049) and 0.25% Trypsin-EDTA solution (Invitrogen, 25002-072). Cells were monitored routinely for mycoplasma contamination using a mycoplasma detection PCR kit (ABM, G238). A FLAG (NiV-F-FLAG) or HA (NiV-F-HA) tag was inserted after residue 104 of codon-optimized NiV-F in pcDNA3.1 vector. NiV-F-YA, L53D, V108D, Q393L, and LI4A were produced by site-directed mutagenesis using the NiV-F-FLAG in pcDNA3.1 vector as a template^{6,10,19}. AP2µ2-mcherry plasmid is a gift from Christien Merrifield (Addgene # 27672). The HA-tagged NiV-G and GFP-tagged NiV-M were constructed previously.

Immunofluorescence (IF) for SMLM

For SMLM imaging on cells, 1×10⁵ PK13 or HeLa cells were seeded on coverslips (Marienfeld #1.5H, 18 mm) coated with 2.5 µg fibronectin (Sigma-Aldrich, F4759-2 mg) in a 12-well plate, and transfected with 1 µg NiV-F variants using lipofectamine 3000 (Invitrogen, L3000015) on the following day. E64d was dissolved in methanol and pitstop2 in DMSO as recommended by the manufacturer. E64d was added to cells at 3 hrs post-transfection for a total treatment time of 15 hrs. Pitstop2 was added to cells at 10 hrs post-transfection for a total treatment time of 8 hrs. At 18 hrs posttransfection, cells were fixed with PBS containing 4% paraformaldehyde (PFA; Electron Microscopy Sciences; 50980487) and 0.2% glutaraldehyde (Sigma-Aldrich, G5882-50ml) for 90 min at room temperature. Cells were treated with signal enhancer image-IT-Fx (Life Technologies, I36933) for 30 min at room temperature, and then blocked using BlockAid (Life Technologies, B10710) for 1 hr at room temperature. The FLAG-tagged NiV-F and mutants were detected by the anti-FLAG mouse monoclonal antibody (Sigma-Aldrich, F1804) and an Alexa Fluor® 647 conjugated donkey anti-mouse secondary antibody (Invitrogen, A31571). Cells were incubated with primary antibody overnight at 4°C, and then with the secondary antibody for 1 hr at room temperature. Each antibody incubation is followed by five PBS washes, 5 min each time. Cells were then fixed in PBS containing 4% PFA for 10 min at room temperature.

SMLM setup, imaging, and data analysis

Imaging was performed on a custom-built SMLM. Briefly, the microscope was built upon an apochromatic TIRF oil-immersion objective lens (Nikon, 60x; numerical aperture 1.49). Four lasers were used for excitation: a 639 nm laser (MRL-FN-639, 500 mW) for exciting Alexa Fluor® 647, a 532 nm laser (MGL-III-532-300 mW) for exciting cy3B, a 488 nm laser (MBL-F-473-300 mW) for exciting GFP, and a 405 nm laser (MDL-III-405-100 mW) for reactivating Alexa Fluor 647 and cy3B. The emission fluorescence was separated using appropriate dichroic mirrors and filters (Semrock) and detected by electron multiplying charge-coupled devices (EMCCD; Ixon, Andor). A feedback loop was employed to control the sample drift to <1 nm laterally and 2.5 nm axially.

Fluorescence beads (Life technologies, F8799) were added to samples as fiducial markers for drift control. Samples were immersed in oxygen-scavenging buffer supplemented with 50 mM mercaptoethylamine (Sigma Aldrich, 30070-10G). The expression level of the protein of interest in individual cells was determined by measuring the average emission fluorescence intensity of an area of $27 \times 27 \mu\text{m}^2$. For SMLM imaging, samples were exposed to a laser power density of 1 kW/cm^2 for the 639 nm lasers to activate Alexa Fluor® 647. A total of 40,000 images were acquired at 50 Hz for the reconstruction of one SMLM image. Custom-written software in MATLAB (Mathworks) was used to reconstruct SMLM images. Clusters of NiV-F localizations on cell surface resolved by SMLM were identified and characterized using ClusDoC. The min points and ϵ were set at 4 and 20, respectively.

VLP production and immunofluorescence

To produce NiV VLPs, HEK293T cells were transfected by NiV-F-FLAG or mutants, NiV-G-HA, and NiV-M-GFP at a 9:1:5 ratio by using polyethylenimine (PEI) at 1 mg/ml (Polysciences, 23966-100). At 48hrs posttransfection, the supernatant of the cell culture was collected and subjected to ultracentrifugation on a 20% sucrose cushion at 125,392 rcf for 90 min. The VLP-containing pellets were resuspended in 5% sucrose-NTE buffer. VLPs were bound to 2.5 μg fibronectin-coated 18 mm coverslips at 37°C for 4 hrs, followed by fixation using PBS containing 4% PFA and 0.2% glutaraldehyde. The immunofluorescence, SMLM setup, and SMLM imaging for cells were followed to stain and image NiV-F-FLAG and mutants on VLPs using SMLM. A widefield image of GFP was acquired and superimposed on the SMLM image of NiV-F. VLPs were identified as GFP-positive particles. The distribution of NiV-F localizations on VLPs was analyzed by using custom-written OPTICS algorithm based on C++ and Point Cloud Library. The classification of the ordered sequence of reachability distances was performed by using custom written one-dimensional convolutional neural network (1D CNN) based on Python and Tensorflow.

SDS-PAGE, and western blot analysis and co-immunoprecipitation

For western blot analysis to confirm the expression of NiV-F-FLAG and mutants, HEK293T cells were seeded in 6-well plate and transfected with 2.5 μg pcDNA3.1 empty vector, NiV-F-FLAG, or mutants. At 28 or 48 hrs posttransfection, cells were lysed in RIPA buffer (Millipore-Sigma, 20-188) supplemented with protease inhibitor (Sigma-Aldrich, 11836170001) on ice for 30min. The cell lysates were collected after centrifuge at $16,000 \times g$ for 20 min at 4°C. The cell lysates were supplemented with 1x SDS loading dye [60 mM Tris-HCl (pH=6.8); 2% SDS; 10% glycerol, 0.025% Brophenol blue] and 15 mM DTT (Thermo Scientific, R0861), and heated at 95°C for denature. The denatured cell lysates were loaded into 10% polyacrylamide gels for SDS-PAGE. Proteins were transferred to PVDF membrane, pore size 0.45 μm (Cytiva, GE10600021).

Membrane was blocked with 1% Bovine serum albumin (Sigma-Aldrich, A9647-50G) in PBS and incubated with anti-FLAG mouse monoclonal antibody (Sigma-Aldrich, F1804). An HRP-conjugated goat anti-mouse secondary antibody and the clarity Western ECL substrate (Biorad, 1705060) were used for protein detection. Images were acquired using Chemic Doc MP Imaging System (BioRad).

For co-immunoprecipitation experiments, HEK293T cells were transfected with the following combinations 1) 2.3 µg empty pcDNA 3.1 vector and 0.2 µg AP2µ2-mcherry, 2) 0.2 µg AP2µ2-mcherry and 2.3 µg NiV-F-FLAG, and 3) 0.2 µg AP2µ2-mcherry and 2.3 µg NiV-F-YA. At 48 hrs post-transfection, cells from 1 well of a 6-well plate were washed with PBS and lysed in 200 µl lysis buffer provided with the µMACS DYKDDDDK isolation kit, and supplemented with protease inhibitors. Cells were isolated on ice for 30 min. Cell debris was removed by centrifuge at 16,000 x g for 20 min at 4°C. 60 µl of cell lysate was set aside for immunoblot analysis, and the rest was used for immunoprecipitation, as recommended by the manufacturer. 6 µl anti-DYKDDDDK microbeads (Miltenyi Biotec, 130-101-591) were added to 140 µl cell lysates and incubated for 30min on ice. µ columns (Miltenyi Biotec, 130-101-591) were prepared according to the manufacturer's instructions. Lysate was run over the columns, and microbeads were washed according to the manufacturer's instruction. 20 µl preheated elution buffer (95°C) was added to the column before eluting the bound immunoprecipitated protein in 50 µl elution buffer. Elute was separated by 10% SDS-PAGE, and proteins were immunoblotted by mouse anti-FLAG and rabbit anti-mcherry primary antibodies. The HRP-conjugated goat anti-mouse and goat anti-rabbit secondary antibodies were used for protein detection.

Flow cytometry

HEK293T cells were seeded in 6-well plate and transfected with 2.5 µg pcDNA3.1 empty vector, NiV-F-FLAG, or mutants. Cells were collected in 1 mL phosphate-buffered saline (PBS) containing 10 mM EDTA after 28hrs post-transfection. The collected cells were incubated on ice for 1 h with primary antibody mouse anti-flag diluted in 1:200. Samples were washed twice in fluorescence-activated cell sorting (FACS) buffer (0.1% FBS with PBS). After washing, samples were incubated with fluorescent anti-mouse Alexa Fluor 647 antibody 1:400 on ice for 45 min. After being washed twice again, samples were read on a flow cytometer (Attune® NxT™ Acoustic Focusing Cytometer, Thermo Fisher) The results were analyzed through FlowJo. The mean fluorescent intensities (MFI) were normalized to the MFIs of the wt NiV F.

Cell-cell fusion

HEK293T cells were seeded in the 12-well plate and transfected with 1µg total DNA of NiV-G and wt-F or mutant-F with a 3:1 ratio using Lipofectamine 3000. Cells were fixed with 4% PFA after 18 hrs post-transfection. Four or more nuclei within a common cytoplasm were considered syncytium. Syncytia were quantified by counting the number of nuclei in the syncytium per 10 × filed (5 fields are counted per group) under microscope TE2000U.

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Author contribution

Conceptualization: Q.W. and Q.L.; Image acquisition: Q.W.; Data analysis and code writing: Q.W., J.L. and Q.L.; Functional analysis: Q.W., Y.L., V.K., G.L.M., J.W.; Resources: Q.L., Supervision: Q.L.; Funding Acquisition: Q.L.. Writing: original draft: Q.W., J.L., and Q.L.; revision: all authors. Competing Interests: The authors declare that they have no competing interests. Code, reagents, and data availability: Custom-written codes are deposited in GitHub or available upon requests. Reagents can be provided by the corresponding author on a completed material transfer agreement. Requests for the reagents and data should be submitted to qian.liu3@mcgill.ca.

Declare of interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

Summary:

In this work by Wang et al., the authors use single-molecule super-resolution microscopy together with biochemical assays to quantify the organization of Nipah virus fusion protein F (NiV-F) on cell and viral membranes. They find that these proteins form nanoscale clusters which favors membrane fusion activation, and that the physical parameters of these clusters are unaffected by protein expression level and endosomal cleavage. Furthermore, they find that the cluster organization is affected by mutations in the trimer interface on the NiV-F ectodomain and the putative oligomerization motif on the transmembrane domain, and that the clusters are stabilized by interactions among NiV-F, the AP2-complex, and the clathrin coat assembly. This work improves our understanding of the NiV fusion machinery, which may have implications also for our understanding of the function of other viruses.

Strengths:

The conclusions of this paper are well-supported by the presented data. This study sheds light on the activation mechanisms underlying the NiV fusion machinery.

Weaknesses:

The authors provide limited details of the convolutional neural network they developed in this work. Even though custom-codes are made available, a description of the network and specifications of how it was used in this work would aid the readers in assessing its performance and applicability. The same holds for the custom-written OPTICS algorithm. Furthermore, limited details are provided for the imaging setup, oxygen scavenging buffer, and analysis for the single-molecule data, which limits reproducibility in other laboratories. The claim of 10 nm resolution is not backed up by data and seems low given the imaging conditions and fluorophores used. Fourier Ring Correlation analysis would have validated this claim. If the authors refer to localization precision rather than resolution, then this should be specified and appropriate data provided to support this claim.

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Reviewer #2 (Public Review):

Summary:

In this manuscript, Wang and co-workers employ single molecule light microscopy (SMLM) to detect Nipah virus Fusion protein (NiV-F) in the surface of cells. They corroborate that these glycoproteins form microclusters (previously seen and characterized together with the NiV-G and Nipah Matrix protein by Liu and co-workers (2018) also with super-resolution light microscopy). Also seen by Liu and coworkers the authors show that the level of expression of NiV-F does not alter the identity of these microclusters nor endosomal cleavage. Moreover, mutations and the transmembrane domain or the hexamer-of-trimer interface seem to have a mild effect on the size of the clusters that the authors quantified. Importantly, it has also been shown that these particles tend to cluster in Nipah VLPs.

Strengths:

The authors have tried to perform SMLM in single VLPs and have shown partially the importance of NiV-F clustering.

Weaknesses:

The labelling strategy for the NiV-F is not sufficiently explained. The use of a FLAG tag in the extracellular domain should be validated and compared with the unlabelled WT NiV-F when expressed in functional pseudoviruses (for example HIV-1 based particles decorated with NiV-F). This experiment should also be carried out for both infection and fusion (including BlaM-Vpr as a readout for fusion). I would also suggest to run a time-of-addition BlaM experiment to understand how this particular labelling strategy affects single virion fusion as compared to the the WT. It would also be very important to compare the FLAG labelling approach with recent advances in the field (for instance incorporating noncanonical amino acids (ncAAs) into NiV-F by amber stop-codon suppression, followed by click chemistry).

The correlation between the existence of microclusters of a particular size and their functionality is missing. Only cell-cell fusion assays are shown in supplementary figures and clearly, single virus entry and fusion cannot be compared with the biophysics of cell-cell fusion. Not only the environment is completely different, membrane curvature and the number of NiV-F drastically varies also. Therefore, specific fusion assays (either single virus tracking and/or time-of-addition BlaM kinetics with functional pseudoviruses) are needed to substantiate this claim.

The authors also claim they could not characterize the number of NiV-F particles per cluster. Another technique such as number and brightness (Digman et al., 2008) could support current SMLM data and identify the number of single molecules per cluster. Also, this

technology does not require complex microscopy apparatus. I suggest they perform either confocal fluorescence fluctuation spectroscopy or TIRF-based nanob to validate the clusters and identify how many molecule are present in these clusters. Also, it is not clear how many cells the authors employ for their statistics (at least 30-50 cells should be employed and not consider the number of events blinking events). I hope the authors are not considering only a single cell to run their stats... The differences between the mutants and the NiV-F is minor even if their statistical analyses give a difference (they should average the number and size of the clusters per cell for a total of 30-50 cells with experiments performed at least in three different cells following the same protocol). They should also compare the level of expression (with the number of molecules per cell provided by number and brightness) with the total number of clusters. Overall, it seems that the authors have only evaluated a very low number of cells.

The same applies to the VLP assay. I assume the authors have only taken VLPs expressing both NiV-M and NiV-F (and NiV-G). But even if this is not clearly stated I would urge the authors to show how many viruses were compared per condition (normally I would expect 300 particles per condition coming from three independent experiments). As a negative control to evaluate the cluster effect I would mix the different conditions. Clearly you have clusters with all conditions and the differences in clustering depending on each condition are minimal. Therefore you need to increase the n for all experiments.

<https://doi.org/10.7554/eLife.97017.1.sa1>

Reviewer #3 (Public Review):

Summary:

The manuscript by Wang and colleagues describes single molecule localization microscopy to quantify the distribution and organization of Nipah virus F expressed on cells and on virus-like particles. Notably the crystal structure of F indicated hexameric assemblies of F trimers. The authors propose that F clustering favors membrane fusion.

Strengths:

The manuscript provides solid data on imaging of F clustering with the main findings of:

- F clusters are independent of expression levels
- Proteolytic cleavage does not affect F clustering
- Mutations that have been reported to affect the hexamer interface reduce clustering on cells and its distribution on VLPs
- F nanoclusters are stabilized by AP

Weaknesses:

The relationship between F clustering and fusion is per se interesting, but looking at F clusters on the plasma membrane does not exclude that F clustering occurs for budding. Many viral glycoproteins cluster at the plasma membrane to generate micro domains for budding. This does not exclude that these clusters include hexamer assemblies or clustering requires hexamer assemblies.

Assuming that the clusters are important for entry, hexameric clusters are not unique to Nipah virus F. Similar hexameric clusters have been described for the HEF on influenza virus C particles (Halldorsson et al 2021) and env organization on Foamy virus particles (Effantin et al 2016), both with specific interactions between trimers. What is the organization of F on Nipah virus particles? If F requires to be hexameric for entry, this should be easily imaged by EM on infectious or inactivated virus particles.

AP stabilization of the F clusters is curious if the clusters are solely required for entry? Virus

entry does not recruit the clathrin machinery. Is it possible that F clusters are endocytosed in the absence of budding?

Other points:

Fig. 3: Some of the V108D and L53D clusters look similar in size than wt clusters. It seems that the interaction is important but not absolutely essential? Would a double mutant abrogate clustering completely?

Fig. 4: The distribution of F on VLPs should be confirmed by cryoEM analyses. This would also confirm the symmetry of the clusters.

The manuscript by Chernomordik et al. JBC 2004 showed that influenza HA outside the direct contact zone affects fusion, which could be further elaborated in the context of F clusters and the fusion mechanism.

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